

SANTIAGO QUINTA NORMAL, Chile

WMO: 855770

Lat: 33.4450S

Lon: 70.6828W

Elev: 520

StdP: 95.23

Time Zone: -4.00 (CHL)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.4	1.5	-3.2	3.1	14.9	-1.8	3.5	11.2	3.2	11.3	2.5	11.6	0.0	N/A	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.8	32.2	17.0	31.0	16.9	29.9	16.7	18.8	28.8	18.2	28.3	17.6	27.7	3.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.8	11.2	22.3	14.2	10.8	21.6	13.6	10.4	21.0	55.4	28.8	53.4	28.3	51.5	27.8	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.3	3.7	3.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	22.2	21.8	19.9	15.5	12.7	9.7	9.3	10.7	13.3	15.7	18.5	20.6
	DBStd	5.06	1.56	1.88	1.91	2.42	2.32	2.26	2.46	2.64	2.65	2.66	2.32	2.02
	HDD10.0	104	0	0	0	0	5	33	43	21	2	0	0	0
	HDD18.3	1319	0	2	7	87	176	261	281	236	151	89	26	5
	CDD10.0	2219	377	331	307	167	88	23	21	44	103	176	256	328
	CDD18.3	391	119	99	56	4	0	0	0	0	1	7	31	74
	CDH23.3	4666	1229	971	675	129	6	0	1	14	46	168	486	939
	CDH26.7	1473	454	330	189	20	0	0	0	2	4	32	137	305
Wind	WSAvg	0.9	1.5	1.3	1.1	0.7	0.5	0.4	0.4	0.6	0.9	1.1	1.3	1.5
Precipitation	PrecAvg	258	0	1	3	11	37	70	57	39	19	10	6	3
	PrecMax	588	7	25	26	98	169	228	247	159	108	53	66	42
	PrecMin	64	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	131	1	4	6	17	37	63	54	35	24	13	12	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.9	33.0	32.3	28.9	24.4	21.3	22.4	25.6	27.1	30.0	32.4	33.1
		MCWB	17.7	17.0	16.6	15.0	13.6	11.7	12.7	13.7	14.1	16.3	16.1	17.4
	2%	DB	32.2	31.5	30.4	26.8	21.9	19.2	19.3	22.1	24.5	27.2	29.9	31.4
		MCWB	17.5	17.2	16.3	14.6	12.8	11.3	11.3	12.2	13.8	14.5	15.7	16.7
	5%	DB	30.8	30.2	28.8	24.7	19.9	17.1	16.8	19.1	22.5	25.1	28.3	29.7
		MCWB	17.4	17.1	16.2	14.1	12.3	10.5	10.2	11.5	13.3	14.1	15.4	16.5
	10%	DB	29.4	28.8	27.3	22.7	18.0	15.0	14.8	16.5	19.7	23.0	26.4	28.4
		MCWB	17.3	17.0	15.9	13.8	12.1	9.9	9.5	10.5	12.2	13.5	15.1	16.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	19.7	18.6	16.5	15.4	13.4	13.4	14.3	15.7	17.3	17.7	18.7
		MCDB	30.2	29.5	28.2	25.2	19.6	18.8	20.2	23.7	24.8	27.1	28.1	29.1
	2%	WB	18.9	18.8	17.6	15.6	14.1	12.2	12.0	12.9	14.6	15.6	16.7	17.9
		MCDB	29.1	28.4	27.4	24.0	18.7	16.8	17.1	20.1	22.7	24.7	27.3	28.6
	5%	WB	18.2	18.2	16.9	14.9	13.4	11.2	11.0	11.9	13.6	14.7	16.0	17.2
		MCDB	28.6	27.6	26.4	22.6	18.0	15.1	15.0	17.9	21.0	23.4	26.5	28.0
	10%	WB	17.6	17.5	16.3	14.1	12.6	10.5	10.2	10.9	12.7	13.8	15.3	16.6
		MCDB	27.9	27.0	25.7	21.5	16.9	14.0	13.8	15.7	18.7	21.9	25.2	27.1
Mean Daily Temperature Range	5% DB	MDBR	14.8	14.7	15.1	13.3	10.9	10.6	10.8	11.2	11.7	13.1	14.5	15.5
		MCDBR	17.1	16.4	17.4	17.3	14.7	15.0	15.0	16.6	17.1	17.4	18.1	17.4
		MCWBR	5.8	5.6	6.0	6.6	6.4	7.6	7.5	7.7	7.1	6.6	6.5	6.4
	5% WB	MCDBR	14.6	14.2	15.5	14.7	11.4	11.9	12.6	14.0	15.0	15.2	15.9	15.9
		MCWBR	5.6	5.4	5.8	6.2	5.4	6.5	6.6	6.8	6.5	6.2	6.3	6.3
Clear-Sky Solar Irradiance	taub	0.345	0.344	0.328	0.324	0.315	0.302	0.304	0.318	0.338	0.329	0.320	0.332	
	taud	2.407	2.423	2.458	2.461	2.484	2.546	2.502	2.451	2.373	2.409	2.447	2.429	
	Ebn at Noon	992	973	952	898	853	843	858	892	925	976	1011	1009	
	Edh at Noon	125	119	107	95	82	72	79	94	114	119	120	123	
All-Sky Solar Radiation	RadAvg	8.63	7.77	6.36	4.41	2.81	2.13	2.33	3.04	4.40	5.95	7.62	8.63	
	RadStd	0.40	0.28	0.34	0.44	0.33	0.30	0.25	0.38	0.47	0.46	0.47	0.31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.38	N/A	-1.17	+0.58	N/A	N/A	N/A	N/A	+137	+93

Nomenclature: See separate page